

ENPC IMI Project – Call No. 5

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April 7, 2026

Agenda

- 1 E-Mail feedback
- 2 Global Comparison
- 3 Snapshots: Standard Stencil
- 4 Snapshots: Diagonal Stencil
- 5 Snapshots: Strang MP5

A Discrepancy in the Diffusion Operator (1/2)

The Contradiction

- Mobile cluster set defined as:

$$M = \{(0, 1), (1, 1)\}$$

- But energy functions given for:

$$E_{(1,0)}(x), \quad E_{(1,1)}(x)$$

Chosen Resolution

Modify set to match energies:

$$M = \{(1, 0), (1, 1)\}$$

Resulting Diffusion Coefficients

$$D_{xx} = \frac{1}{2}(1)^2 d(x) + \frac{1}{2}(1)^2 d(x) = d(x)$$

$$D_{yy} = \frac{1}{2}(0)^2 d(x) + \frac{1}{2}(1)^2 d(x) = \frac{1}{2}d(x)$$

$$D_{xy} = (1)(0)d(x) + (1)(1)d(x) = d(x)$$

A Discrepancy in the Diffusion Operator (2/2)

Since yesterday's mail:

- Keep original set:

$$M = \{(0, 1), (1, 1)\}$$

- Assume energies:

$$E_{(0,1)}(x), \quad E_{(1,1)}(x)$$

Please note, that we **do not** test this currently:

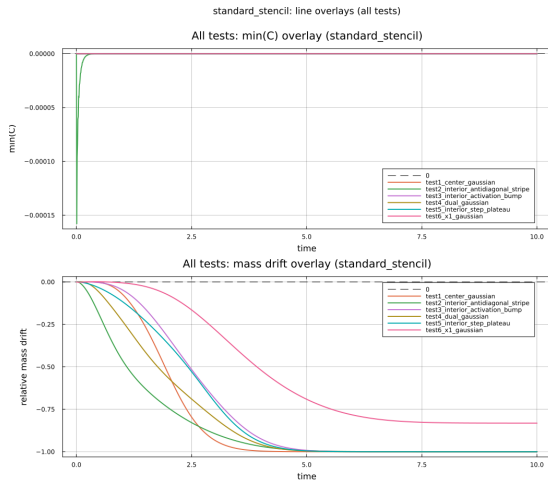
$$m_{(1,0)} = \iint C(x,y)x \, dx dy \quad (1)$$

$$m_{(0,1)} = \iint C(x,y)y \, dx dy. \quad (2)$$

But we test for general mass conservation right now:

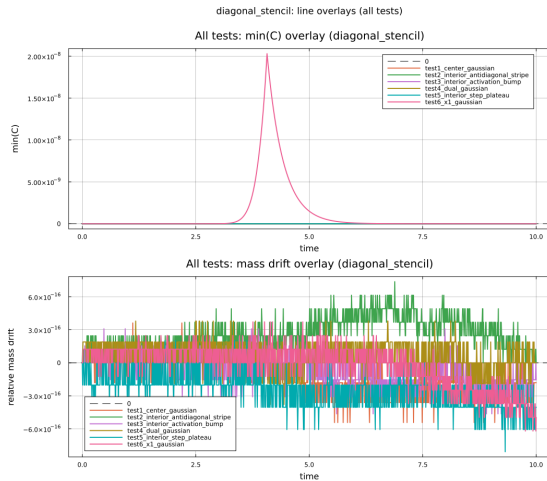
$$m = \iint C(x,y) \, dx dy \quad (3)$$

All Tests – Standard Stencil



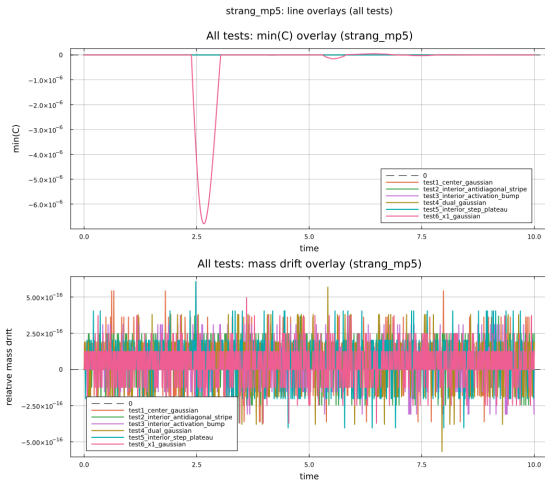
Top: $\min(C)$ overlay. Bottom: relative mass-drift overlay.

All Tests – Diagonal Stencil



Top: min(C) overlay. Bottom: relative mass-drift overlay.

All Tests – Strang MP5

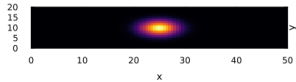


Top: min(C) overlay. Bottom: relative mass-drift overlay.

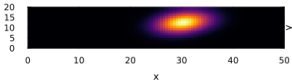
Test 1 – Center Gaussian – Standard

test1_center_gaussian snapshots (local scale per panel, no colorbar)

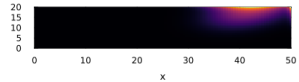
t = 0.000 | min=6.47e-26 max=1.00e+00



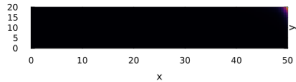
t = 0.501 | min=6.47e-26 max=4.98e-01



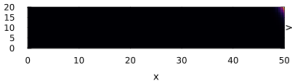
t = 1.502 | min=6.47e-26 max=5.44e-01



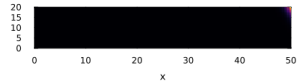
t = 3.498 | min=1.22e-27 max=2.25e-01



t = 6.502 | min=2.71e-31 max=2.36e-05



t = 10.000 | min=1.52e-35 max=2.59e-10



Concentration snapshots over time with local per-panel color scaling.

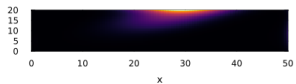
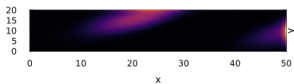
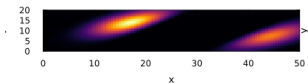
Test 2 – Interior Antidiagonal Stripe – Standard

test2_interior_antidiagonal_stripe snapshots (local scale per panel, no colorbar)

t = 0.000 | min=0.00e+00 max=2.22e-01

t = 0.501 | min=-1.33e-07 max=3.60e-01

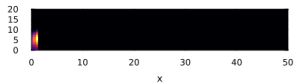
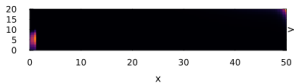
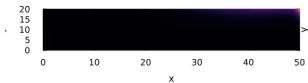
t = 1.502 | min=-5.14e-10 max=1.95e-01



t = 3.498 | min=0.00e+00 max=1.20e-01

t = 6.502 | min=0.00e+00 max=2.14e-03

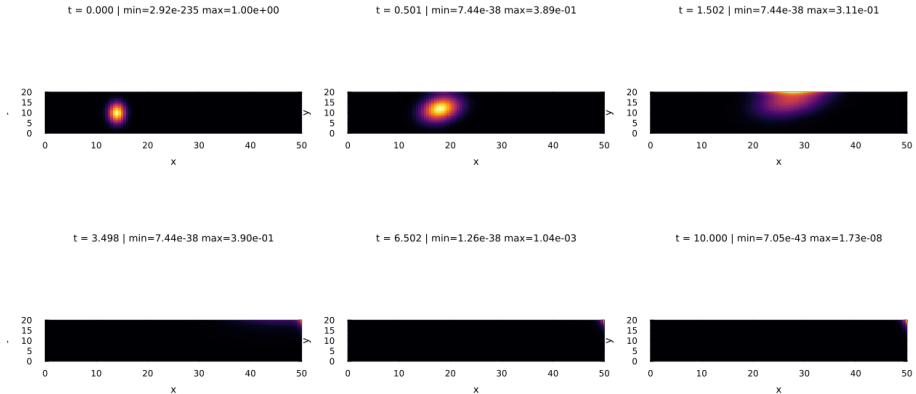
t = 10.000 | min=0.00e+00 max=1.47e-03



Concentration snapshots over time with local per-panel color scaling.

Test 3 – Interior Activation Bump – Standard

test3_interior_activation_bump snapshots (local scale per panel, no colorbar)

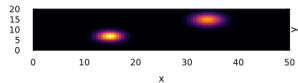


Concentration snapshots over time with local per-panel color scaling.

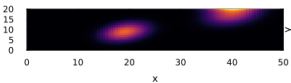
Test 4 – Dual Gaussian – Standard

test4_dual_gaussian snapshots (local scale per panel, no colorbar)

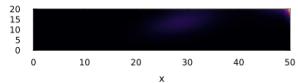
t = 0.000 | min=1.58e-31 max=1.00e+00



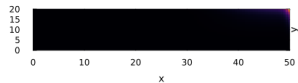
t = 0.501 | min=-1.98e-11 max=5.50e-01



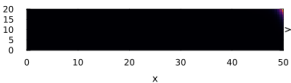
t = 1.502 | min=-7.00e-13 max=1.11e+00



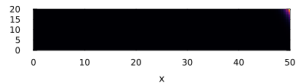
t = 3.498 | min=1.95e-32 max=4.85e-01



t = 6.502 | min=4.33e-36 max=7.42e-04



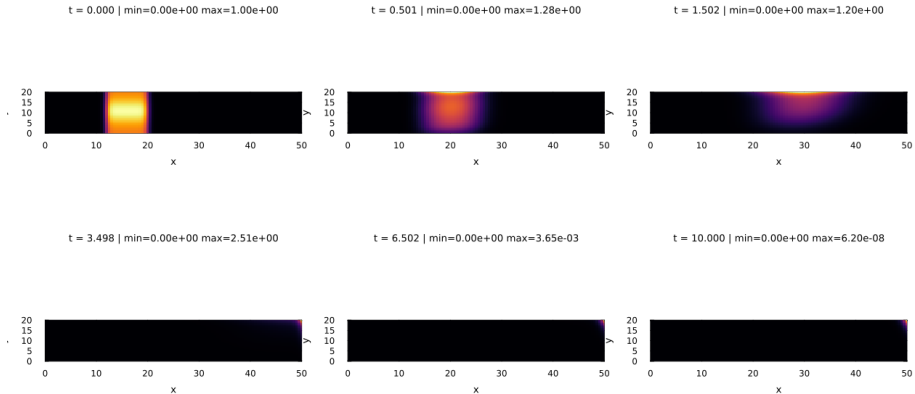
t = 10.000 | min=2.43e-40 max=1.12e-08



Concentration snapshots over time with local per-panel color scaling.

Test 5 – Interior Step Plateau – Standard

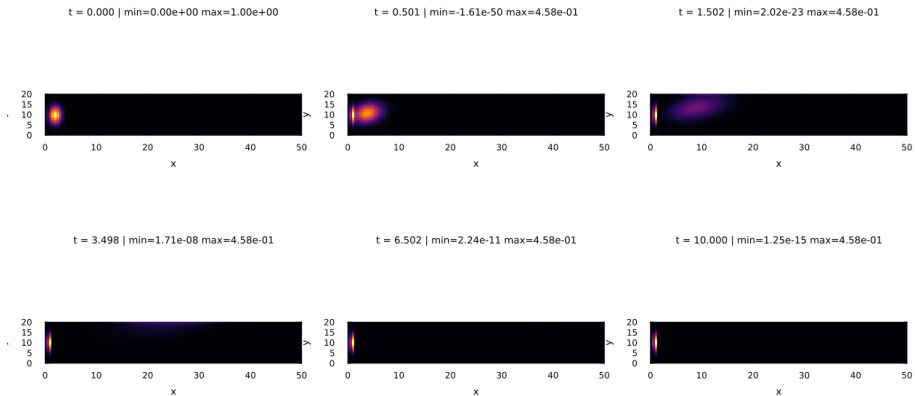
test5_interior_step_plateau snapshots (local scale per panel, no colorbar)



Concentration snapshots over time with local per-panel color scaling.

Test 6 – x1 Gaussian – Standard

test6_x1_gaussian snapshots (local scale per panel, no colorbar)

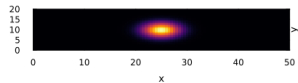


Concentration snapshots over time with local per-panel color scaling.

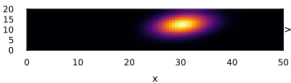
Test 1 – Center Gaussian – Diagonal

test1_center_gaussian snapshots (local scale per panel, no colorbar)

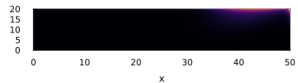
t = 0.000 | min=6.47e-26 max=1.00e+00



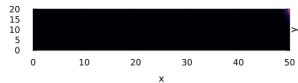
t = 0.501 | min=6.47e-26 max=4.99e-01



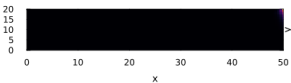
t = 1.502 | min=6.47e-26 max=1.88e+00



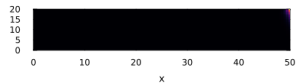
t = 3.498 | min=1.22e-27 max=2.21e+01



t = 6.502 | min=2.71e-31 max=2.09e+01



t = 10.000 | min=1.52e-35 max=2.08e+01



Concentration snapshots over time with local per-panel color scaling.

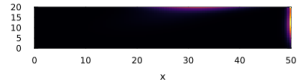
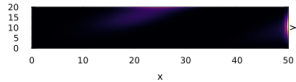
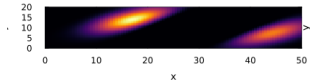
Test 2 – Interior Antidiagonal Stripe – Diagonal

test2_interior_antidiagonal_stripe snapshots (local scale per panel, no colorbar)

t = 0.000 | min=0.00e+00 max=2.22e-01

t = 0.501 | min=0.00e+00 max=1.01e+00

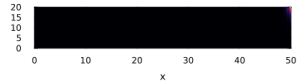
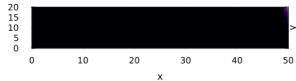
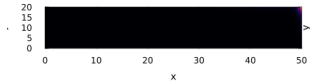
t = 1.502 | min=0.00e+00 max=1.23e+00



t = 3.498 | min=0.00e+00 max=1.06e+01

t = 6.502 | min=0.00e+00 max=1.50e+01

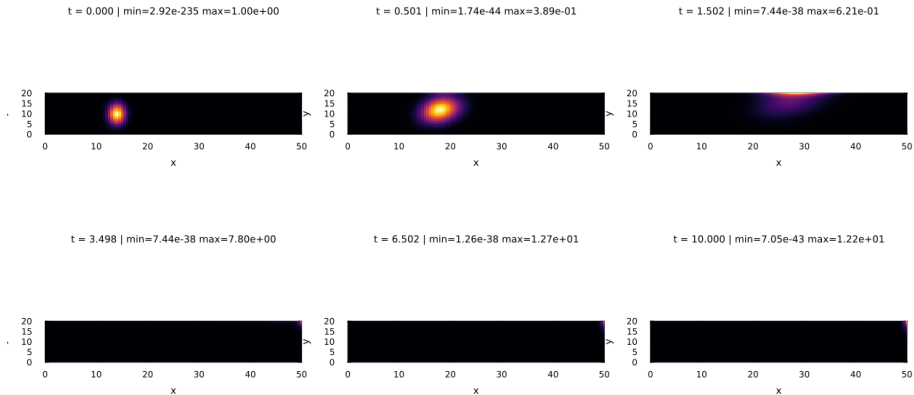
t = 10.000 | min=0.00e+00 max=1.51e+01



Concentration snapshots over time with local per-panel color scaling.

Test 3 – Interior Activation Bump – Diagonal

test3_interior_activation_bump snapshots (local scale per panel, no colorbar)

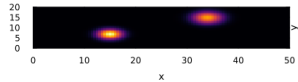


Concentration snapshots over time with local per-panel color scaling.

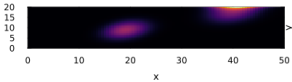
Test 4 – Dual Gaussian – Diagonal

test4_dual_gaussian snapshots (local scale per panel, no colorbar)

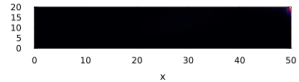
t = 0.000 | min=1.58e-31 max=1.00e+00



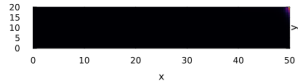
t = 0.501 | min=1.58e-31 max=8.61e-01



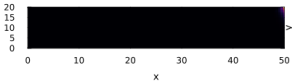
t = 1.502 | min=1.58e-31 max=9.97e+00



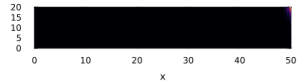
t = 3.498 | min=1.95e-32 max=1.84e+01



t = 6.502 | min=4.33e-36 max=2.02e+01



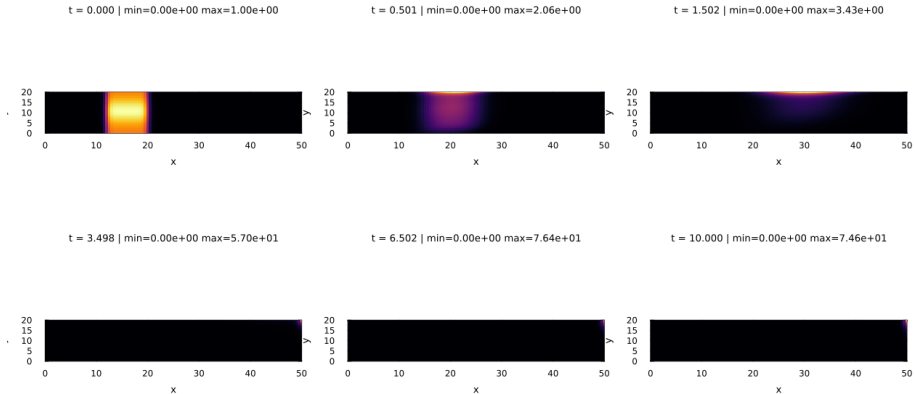
t = 10.000 | min=2.43e-40 max=1.99e+01



Concentration snapshots over time with local per-panel color scaling.

Test 5 – Interior Step Plateau – Diagonal

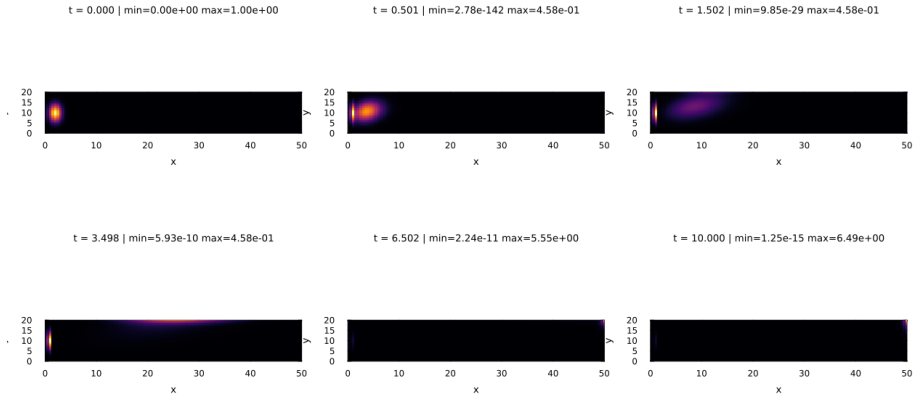
test5_interior_step_plateau snapshots (local scale per panel, no colorbar)



Concentration snapshots over time with local per-panel color scaling.

Test 6 – x1 Gaussian – Diagonal

test6_x1_gaussian snapshots (local scale per panel, no colorbar)

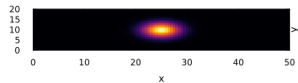


Concentration snapshots over time with local per-panel color scaling.

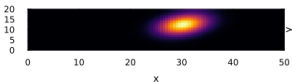
Test 1 – Center Gaussian – Strang MP5

test1_center_gaussian snapshots (local scale per panel, no colorbar)

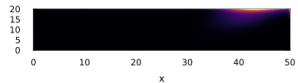
t = 0.000 | min=6.47e-26 max=1.00e+00



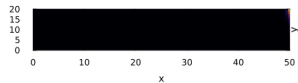
t = 0.501 | min=6.47e-26 max=5.91e-01



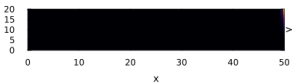
t = 1.502 | min=6.47e-26 max=1.59e+00



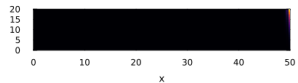
t = 3.498 | min=3.57e-26 max=2.00e+01



t = 6.502 | min=-8.41e-27 max=1.47e+01



t = 10.000 | min=-1.98e-28 max=1.31e+01



Concentration snapshots over time with local per-panel color scaling.

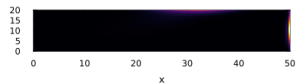
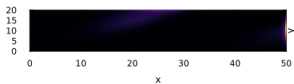
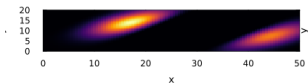
Test 2 – Interior Antidiagonal Stripe – Strang MP5

test2_interior_antidiagonal_stripe snapshots (local scale per panel, no colorbar)

t = 0.000 | min=0.00e+00 max=2.22e-01

t = 0.501 | min=0.00e+00 max=1.38e+00

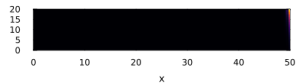
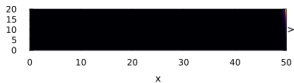
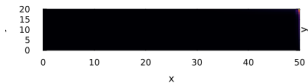
t = 1.502 | min=0.00e+00 max=1.69e+00



t = 3.498 | min=0.00e+00 max=8.98e+00

t = 6.502 | min=0.00e+00 max=9.81e+00

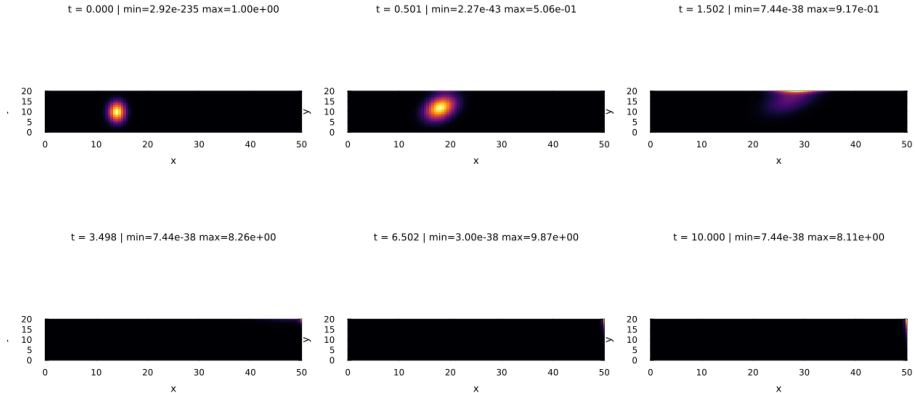
t = 10.000 | min=-5.17e-12 max=8.85e+00



Concentration snapshots over time with local per-panel color scaling.

Test 3 – Interior Activation Bump – Strang MP5

test3_interior_activation_bump snapshots (local scale per panel, no colorbar)

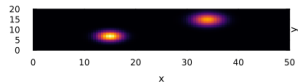


Concentration snapshots over time with local per-panel color scaling.

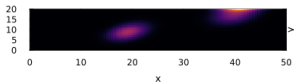
Test 4 – Dual Gaussian – Strang MP5

test4_dual_gaussian snapshots (local scale per panel, no colorbar)

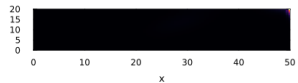
t = 0.000 | min=1.58e-31 max=1.00e+00



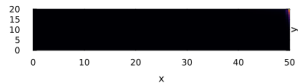
t = 0.501 | min=1.58e-31 max=1.06e+00



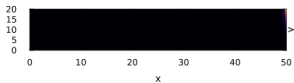
t = 1.502 | min=1.58e-31 max=1.25e+01



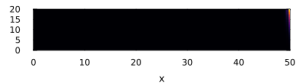
t = 3.498 | min=9.45e-33 max=1.83e+01



t = 6.502 | min=-1.58e-20 max=1.46e+01



t = 10.000 | min=-1.92e-23 max=1.27e+01

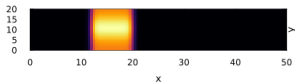


Concentration snapshots over time with local per-panel color scaling.

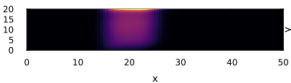
Test 5 – Interior Step Plateau – Strang MP5

test5_interior_step_plateau snapshots (local scale per panel, no colorbar)

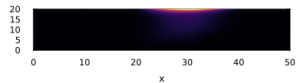
t = 0.000 | min=0.00e+00 max=1.00e+00



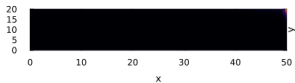
t = 0.501 | min=0.00e+00 max=2.24e+00



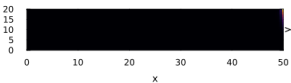
t = 1.502 | min=0.00e+00 max=4.36e+00



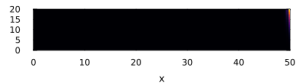
t = 3.498 | min=0.00e+00 max=6.38e+01



t = 6.502 | min=-1.89e-20 max=5.78e+01



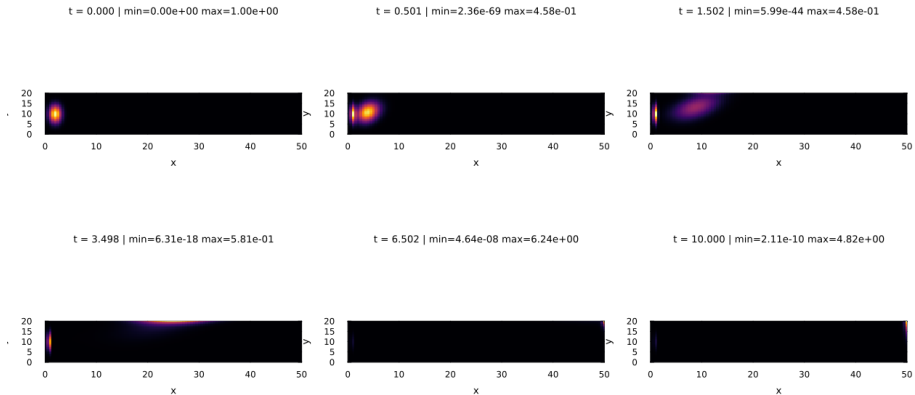
t = 10.000 | min=-5.11e-23 max=4.87e+01



Concentration snapshots over time with local per-panel color scaling.

Test 6 – x1 Gaussian – Strang MP5

test6_x1_gaussian snapshots (local scale per panel, no colorbar)



Concentration snapshots over time with local per-panel color scaling.